COMPLETING THE SQUARE

Higher Tier

Completed-square form, turning points and graph applications

Name .....

Date .....

11 questions	37 marks	2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working. Give exact values where required.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Start with reading completed-square form, then practise rewriting expressions, finding turning points and sketching graphs. The final questions combine these skills with other algebra.

The original exam question numbers and part labels are retained. Use the practice numbers and index to navigate. Practice 04 includes only part (a) of the original question.

This is an independently compiled practice resource, not an official Pearson examination paper. Questions are selected from standard UK June and November papers, 2017-2025. No solutions are included.

Question index

Higher Tier | 11 questions | 37 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	November 2020	1MA1/2H	Q17	1	3
02	November 2017	1MA1/3H	Q13	2	4
03	June 2019	1MA1/1H	Q19(i)-(ii)	3	5
04	November 2023	1MA1/3H	Q17(a)(i)-(ii)	3	6
05	June 2024	1MA1/2H	Q20	3	7
06	November 2022	1MA1/3H	Q21	3	8
07	November 2021	1MA1/1H	Q22	4	9
08	November 2025	1MA1/1H	Q17	4	10
09	November 2018	1MA1/1H	Q17	4	11
10	November 2019	1MA1/1H	Q21	5	12
11	November 2024	1MA1/3H	Q23	5	13

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

Practice 04: only Q17(a)(i)-(ii) is included, worth 3 marks.

Practice 01 is a short warm-up using an expression already in squared form.

17 Write down the coordinates of the turning point on the graph of $y = (x + 12)^2 - 7$

(..... , ...)

(Total for Question 17 is 1 mark)

Additional working space

13 Write $x^2 + 6x - 7$ in the form $(x + a)^2 + b$ where a and b are integers.

.....

(Total for Question 13 is 2 marks)

Additional working space

19 Given that $x^2 - 6x + 1 = (x - a)^2 - b$ for all values of x,

(i) find the value of a and the value of b.

$a =$ .....

$b =$ .....

(2)

(ii) Hence write down the coordinates of the turning point on the graph of $y = x^2 - 6x + 1$

(..... , ...)

(1)

(Total for Question 19 is 3 marks)

Additional working space

17 (a) (i) Write $x^2 - 8x + 3$ in the form $(x - a)^2 - b$ where a and b are integers.

.....

(2)

(ii) Hence, write down the coordinates of the turning point on the graph of $y = x^2 - 8x + 3$

(..... , ...)

(1)

Additional working space

20 The curve **C** has equation $y = 2x^2 - 12x + 7$

Find the coordinates of the turning point on **C**.

(..... , ...)

(Total for Question 20 is 3 marks)

Additional working space

21 The equation of a curve is $y = 4x^2 - 56x$
The curve has one turning point.

By completing the square, show that the coordinates of the turning point are $(7, -196)$
You must show all your working.

(Total for Question 21 is 3 marks)

Additional working space

22 Find the coordinates of the turning point on the curve with equation $y = 9 + 18x - 3x^2$
You must show all your working.

(..... , ...)

(Total for Question 22 is 4 marks)

Additional working space

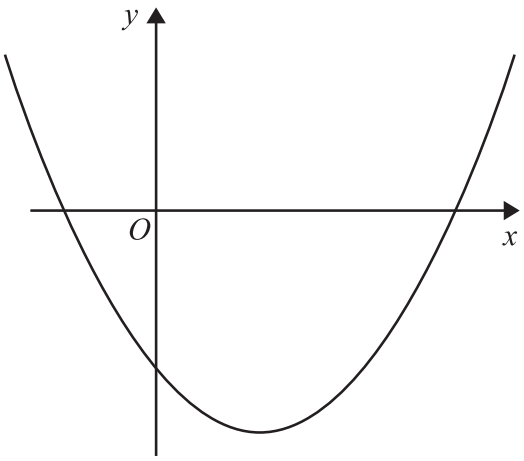
17 Find the coordinates of the turning point of the graph of $y = 5x^2 + 4x + 9$

(..... , ...)

(Total for Question 17 is 4 marks)

Additional working space

17 Here is a sketch of a curve.



The equation of the curve is $y = x^2 + ax + b$ where a and b are integers.

The points $(0, -5)$ and $(5, 0)$ lie on the curve.

Find the coordinates of the turning point of the curve.

(..... , ...)

(Total for Question 17 is 4 marks)

21 Sketch the graph of

$$y = 2x^2 - 8x - 5$$

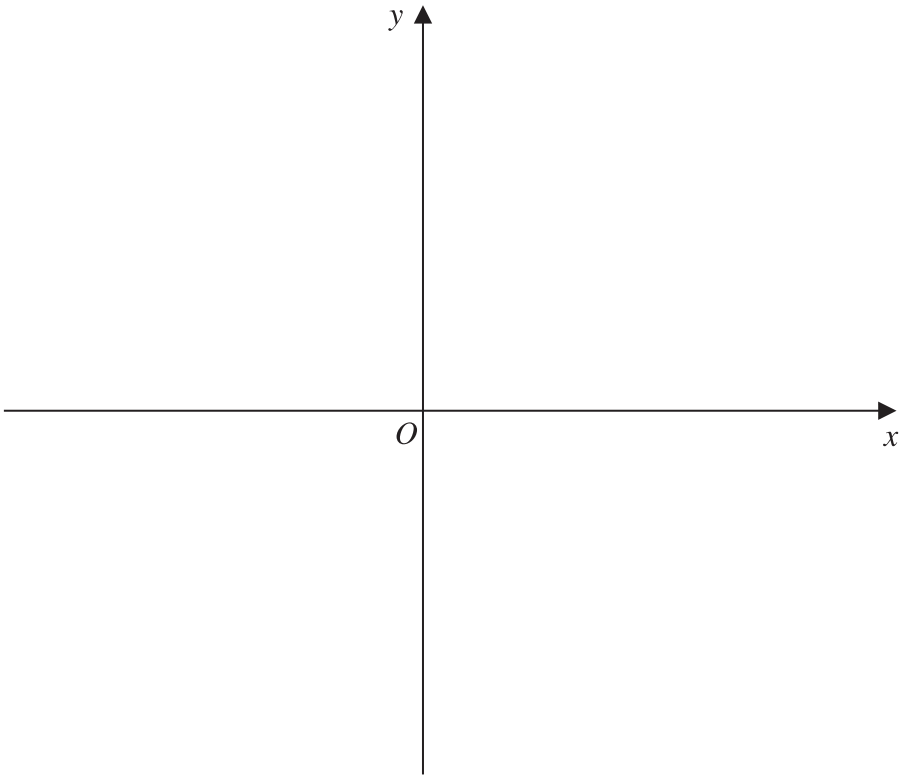
showing the coordinates of the turning point and the exact coordinates of any intercepts with the coordinate axes.

(Total for Question 21 is 5 marks)

23 Sketch the graph of

$$y = x^2 - 6px - 7 \quad \text{where} \quad p > 0$$

showing the coordinates of the turning point, in terms of p, and the coordinates of the intercept with the y-axis.
You must show all your working.



(Total for Question 23 is 5 marks)